

Wei Xu

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EDUCATION

California Institute of Technology

Ph.D. Student in Applied Physics; M.S. in Applied Physics (conferred June 2026)

September 2025 – Present

Pasadena, CA

Tsinghua University

B.S. in Mathematics and Physics, GPA 3.87/4.00

August 2021 – June 2025

Beijing, China

RESEARCH FOCUS

Integrated photonic devices; microresonator frequency combs; optical frequency division; ultra-low-noise microwave synthesis; and precision timing.

RESEARCH PUBLICATIONS

Family of magnetic field-boosted superconductors in rhombohedral graphene.

J. Seo, A. A. Cotten, S. Ye, M. Xu, O. S. Sedeh, H. Weldeyesus, T. Han, Z. Lu, Z. Wu, **W. Xu**, J. Yang, E. Aitken, P. P. Liang, P. Pattanakanvijit, Z. Hadjri, R. Gazizulin, K. Watanabe, T. Taniguchi, M. Li, D. M. Zumbühl, and L. Ju.

Nature, 10.1038/s41586-026-10815-x (2026).

Signatures of chiral superconductivity in rhombohedral graphene.

T. Han, Z. Lu, Z. Hadjri, L. Shi, Z. Wu, **W. Xu**, Y. Yao, A. A. Cotten, O. S. Sedeh, H. Weldeyesus, J. Yang, J. Seo, S. Ye, M. Zhou, H. Liu, G. Shi, Z. Hua, K. Watanabe, T. Taniguchi, P. Xiong, D. M. Zumbühl, L. Fu, and L. Ju.

Nature, 643, 654–661 (2025).

Ballistic PbTe nanowire devices.

Y. Wang, F. Chen, W. Song, Z. Geng, Z. Yu, L. Yang, Y. Gao, R. Li, S. Yang, W. Miao, **W. Xu**, Z. Wang, Z. Xia, H.-D. Song, X. Feng, T. Wang, Y. Zang, L. Li, R. Shang, Q. Xue, K. He, and H. Zhang.

Nano Letters, 23, 11137–11144 (2023).

RESEARCH EXPERIENCE

California Institute of Technology

September 2025 - Present

Graduate Researcher

- **Research Rotations (September 2025 - March 2026):** Conducted laboratory research across multiple Caltech groups.
- **Coupled Optical Parametric Oscillators (December 2025 - March 2026; Advisor: Prof. Alireza Marandi):** Built and debugged a quasi-continuous-wave optical pump path in thin-film lithium-niobate photonic chips and observed optical parametric oscillation on chip.
- **Integrated Photonics, Optical Frequency Combs & Microwave Synthesis (April 2026 - Present; Advisor: Prof. Kerry Vahala):** Design, simulate, lay out, nanofabricate, and characterize integrated photonic devices and microresonator chips for octave-spanning frequency-comb generation and ultra-low-noise microwave synthesis.
- Develop auxiliary-ring microresonator architectures for on-chip f-2f self-referencing and perform broadband optical and microwave characterization for compact precision-timing systems.

Superconductivity in Rhombohedral Graphene & Quantum Capacitance

June 2024 - March 2025

Visiting Student Researcher, MIT (Advisor: Prof. Long Ju)

- Fabricated high-quality, dual-graphite-gated rhombohedral graphene devices, contributing to research on chiral superconductivity published in *Nature*.
- Developed an ultralow-temperature quantum-capacitance measurement setup and an on-chip bilayer-graphene cryogenic amplifier for characterizing strongly correlated electronic states.

AWARDS

Tsinghua University **Overseas Academic Research Support Program** (2024); Beijing Natural Science Foundation ‘**Qiyang**’ Program Fellowship (2023); Tsinghua University **Comprehensive Excellence Scholarship** (2023); **Spark Scientific and Technological Innovation Fellowship**, top 1% at Tsinghua (2023); **Chi-Sun Yeh Physics Academy Program Scholarship** (2021–2024).